

ActiveWAM

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Preserve the evidence. Use the history. Acquire the next view.

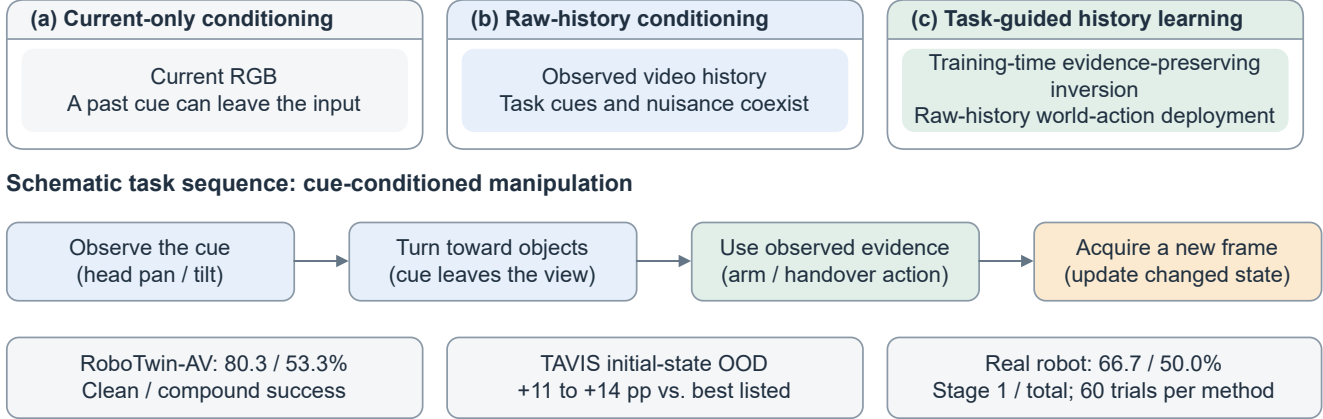


Fig. 1: ActiveWAM connects task-selective history learning with physical observation control. Current-only conditioning, raw history, and task-guided learning differ in their access to a briefly observed cue. Training-time inversion constrains source evidence and its changes; deployment generates head and arm actions from raw history. The sequence is schematic. Numerical panels summarize benchmark and kitchen-task results; physical evaluation uses 20 attempts per task and 60 per method.

Abstract—Reliable manipulation depends on what a robot observes as well as how it acts. Active cameras can expose useful object–hand and object–goal relations, but changing the view also changes the visual context available for control. Recent observations contain both task evidence and distracting appearance variation; suppressing the latter must preserve small cues and visible state changes. We present ActiveWAM, a world–action model that couples task-selective learning from recent visual history with physical observation control. During training, a frozen video prior transforms observed histories under source-evidence and temporal-change constraints. The policy learns from raw and transformed histories with the same future and action targets. At deployment, it uses raw observations to jointly generate head and bimanual actions, updating its finite visual context from new camera exposures. We also introduce RoboTwin-AV, a 50-task extension of RoboTwin with executable pan/tilt control and automatically generated demonstrations. On the independent TAVIS benchmark, ActiveWAM improves initial-state OOD success by 11–14 percentage points over the strongest listed baseline across four robot–suite settings. It achieves 53.3% compound-shift success on RoboTwin-AV and 50.0% total-task success across three composed physical kitchen tasks, exceeding Fast-WAM by 20.3 and 26.7 points, respectively. These results support learning robust visual conditions together with useful observation actions.

I. INTRODUCTION

General-purpose manipulation requires robots to acquire useful visual evidence as well as act on it. Vision–Language–Action (VLA) policies connect observations and instructions to commands [1], [2], [3], while World–Action Models (WAMs) additionally learn the visual consequences of those commands [4], [5], [6]. For bimanual manipulation, co-ordinated actions change the scene while camera actions determine which details the robot can observe.

Useful observation is not equivalent to removing occlusion. During an approach, a view should expose the relation between an object and the end-effector; during placement, the destination becomes equally important. A robot may inspect a cue, turn toward a grasp, and later need to check whether a new event has invalidated the cue. Even in an unobstructed workspace, task-centered framing can make action-relevant details easier to use. Conversely, moving an already well-positioned camera can discard context, introduce image motion, and delay execution. The appropriate observation behavior therefore includes retaining a good view, not only searching for a better one.

Active perception treats observation as a controllable process [7], [8]. AV-ALOHA evaluates active, static, and wrist cameras in bimanual tasks [9]; AVR combines viewpoint and focal-length control for precise manipulation [10]. TAVIS studies conditional selection, search, waiting, and exploratory wrist motion, showing that the benefit of an active camera depends on the task [11]. See2Act couples action refinement to target-centric physical views [12]. These results motivate observation control beyond a universal object-centering rule. They also expose a temporal problem: changing the view can acquire new evidence while removing previously useful evidence from the current image.

Pretrained visual representations support robot transfer [13], [14], while domain randomization targets visual shifts [15]. Yet history still mixes task evidence with appearance, lighting, and camera motion. Suppressing variation can erase a target-selecting color cue or a state change that determines when to act. Stem-OB uses diffusion inversion to improve visual generalization while retaining raw-image

deployment [16]. However, a generic image transform neither specifies which historical evidence must survive nor preserves the temporal changes that can overturn an earlier decision. A video prior must therefore be constrained by the observed source, rather than encouraged to make every frame look similar.

We address the joint question: *how can a world-action model learn to preserve task-relevant evidence in recent observations and control the evidence it acquires next?* Our key insight is to couple a task-selective training transformation with executable observation actions through the same demonstrated continuation. The transformation changes the representation of an observed history; head and arm actions change the physical observation stream. Neither operation substitutes for the other. In particular, transforming a history cannot reveal an unobserved surface, and moving a camera cannot by itself ensure that a previously seen cue remains available to the policy.

We present ActiveWAM, which combines task-guided history inversion with unified future-video and head-arm generation (Fig. 1). Training-Time Inversion (TTI) uses a frozen video prior constrained by task-bearing source evidence and visible temporal changes. A view-aware history adapter represents camera identity, pose, exposure time, and validity alongside the current RGB observation. The policy learns from raw and transformed conditions with identical original future and action targets. At deployment, it uses raw observations to generate one joint trajectory and updates its history after executing a short prefix. It uses neither online inversion nor candidate ranking, and requires no manually annotated optimal viewpoints or information-gain objective.

To evaluate this coupling at bimanual-task scale, we introduce *RoboTwin-AV*, an active-view benchmark extension of RoboTwin 2.0 [17]. It retains all 50 task definitions and native success predicates while adding executable fixed-focal-length pan/tilt control, synchronized RGB observations, and automatically generated head-arm demonstrations. This setting matches a fixed-base dual-arm robot without an extra camera arm or optical zoom. We evaluate the method on the independently released TAVIS benchmark and use RoboTwin-AV for large-scale controlled comparisons. Fixed-camera generalization tests and physical trials complement these active-vision evaluations, separating task success from the costs of obtaining a view.

Our contributions are threefold:

- We introduce task-guided inversion of observed video histories, preserving task evidence and its visible changes within a finite window, while transferring the resulting training signal to a raw-input policy.
- We develop a unified active-view WAM that jointly generates future video and executable head-arm actions from view-aware history, linking evidence retention to physical observation control.
- We introduce RoboTwin-AV, a 50-task benchmark extension with 100 generated training trajectories per task, and evaluate the method through external-baseline comparisons, controlled ablations, and real bimanual

manipulation.

II. RELATED WORK

A. Active Observation and Camera Control

Active-vision policies control sensing to reduce perceptual ambiguity. AV-ALOHA studies camera placement in bimanual manipulation [9]; AVR combines viewpoint and focal-length control [10]. Sun et al. combine Diffusion Policy with look-at inverse kinematics, reducing the orientation decisions learned by the policy [18]. TAVIS evaluates task-dependent gaze, waiting, search, and initial-pose shifts using Head and Hands embodiments [11]. See2Act couples action denoising to test-time viewpoint refinement [12]. Our fixed-focal head has only pan and tilt; observation and manipulation share one action stream, including stay. We compare target centering and object-goal framing because head motion alone does not establish useful sensing.

B. World-Action Models and Unified Generation

Generalist policies scale robot learning through shared transformers [19], [20] and heterogeneous co-training [21]. Visual Foresight plans with learned video prediction [22]; WAMs also generate actions. DreamZero jointly generates video and actions [23]; DiT4DiT conditions an action expert on video-denoising features [24]. EasyWAM-Unified supplies our joint interface [6]. Fast-WAM separates video co-training from action-only inference [4], whereas Cosmos Policy also uses future-state/value predictions for planning [5]. ST-WAM uses semantic futures and history retrieval; GIFT supervises control-relevant intermediate structure [25], [26]. We couple task-selective history transformation to physical sensing actions within one conditional generator. Video co-training and test-time future generation require separate controls.

C. Observation Inversion and Temporal Consistency

DDIM establishes deterministic diffusion trajectories [27]; Null-text Inversion optimizes reconstruction around an inversion trajectory [28]. Prompt-to-Prompt controls text-image cross-attention [29]. Video-P2P, FateZero, and TokenFlow extend source-guided editing through attention control and inter-frame feature correspondences [30], [31], [32]. RF-Solver improves rectified-flow inversion accuracy [33]. In robotics, Stem-OB transfers transformed-input training benefits to raw-image deployment [16]. We constrain a partial history transform by task evidence and visible changes, retaining original continuation targets. Shared-prior, codec, and feature-supervision controls test simpler alternatives; independent source labels assess cue retention beyond agreement with prior features.

III. METHOD

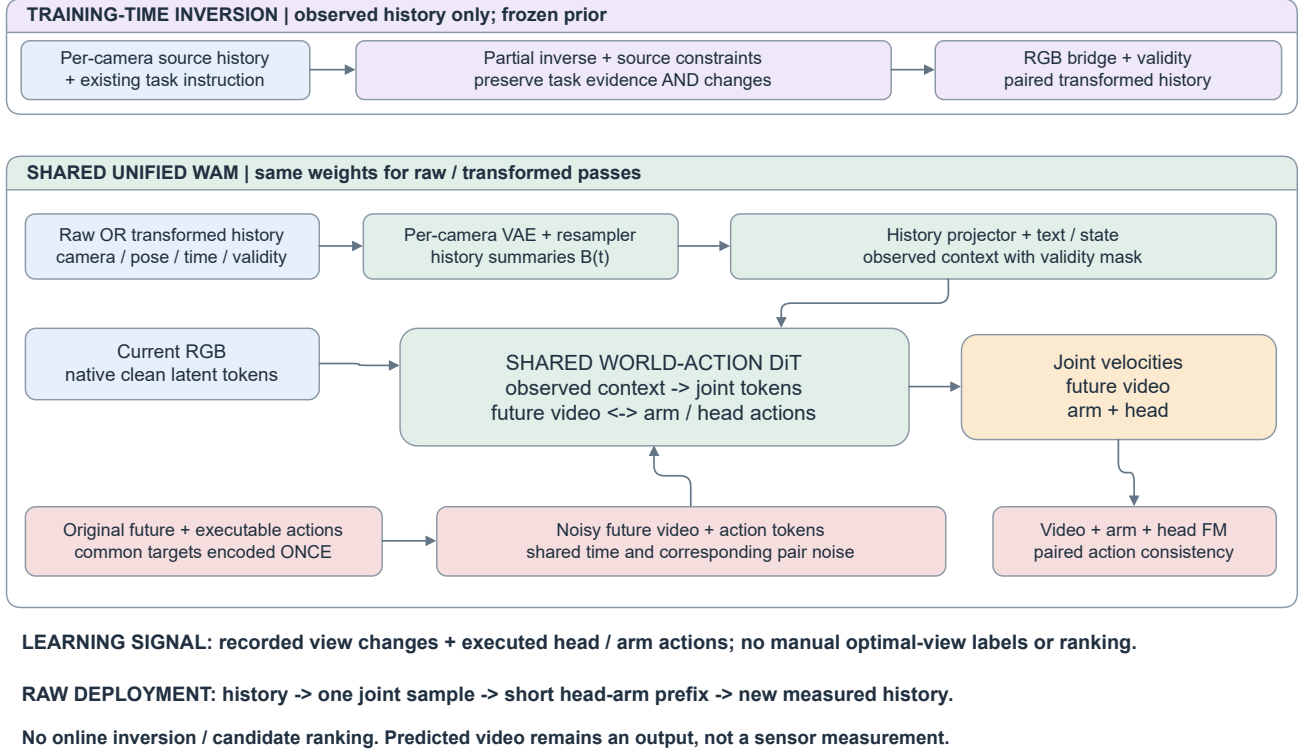


Fig. 2: ActiveWAM overview. Training-time source-constrained inversion transforms observed histories. Shared WAM passes receive raw or transformed conditions, with identical original future/action targets. View-aware history and current RGB condition joint future, arm, and head generation. Deployment retains raw inputs and updates history from new physical exposures after executing a short action prefix.

A. Problem Formulation and Overview

A manipulation policy changes both the scene and its subsequent observations. We preserve task-relevant recent history during learning and jointly generate observation and manipulation actions.

For camera c , let $\mathcal{H}_t^c = (I_{t_1}^c, \dots, I_{t_L}^c)$ contain only frames exposed by decision time t . The condition is $C_t = (\ell, \mathcal{H}_t, s_t, p_{\leq t}, \kappa, \delta_{\leq t})$: instruction, RGB observation history, robot state, measured camera poses, calibration, and acquisition metadata. The default window contains nine frames spanning 0.8 s of exposure time, without recurrent state across windows. Image age and validity distinguish asynchronous or outdated observations. Our fixed-base robot has a single RGB camera on a two-DoF pan/tilt head at fixed focal length. Its action is

$$a_t = [\Delta x_t^l, \Delta \omega_t^l, g_t^l, \Delta x_t^r, \Delta \omega_t^r, g_t^r, \Delta p_t] \in \mathbb{R}^{16}, \quad (1)$$

with six pose increments and a gripper command per arm, and two head increments Δp_t . Zero head motion is a valid *stay* action. Native benchmarks retain their original schemas. We model one conditional distribution,

$$p_\theta(Y_{t+1:t+K}, A_{t:t+H_a-1} \mid C_t), \quad (2)$$

over view-specific future-video latents Y and executable action chunks A . Head rotation changes the field of view and task framing. No optical zoom is used. Approach favors object-hand framing, placement object-goal coverage, and

waiting a stable cue view. Pan/tilt can reacquire off-screen evidence, but not a cue occluded along every reachable viewing direction.

a) Overview: Fig. 2 connects three mechanisms. Training-time task-guided history inversion preserves source evidence and its changes (Sec. III-B). A view-aware history adapter conditions a shared world-action generator on available evidence (Sec. III-C). Paired learning anchors raw and transformed conditions to identical original future/action targets (Sec. III-D). Deployment uses raw history, one joint sample, and actual new measurements after a short executed prefix.

B. Task-Guided History Inversion

Observation inversion supports transformed-input training and raw-input deployment [16]. We extend this principle to *observed video history*. A frozen Wan T2V video prior processes each camera separately. Cameras are not temporal neighbors, and the prior receives neither future observations nor hidden task answers. Bidirectional processing within $\mathcal{H}_t$ is causal for decision t because every frame is already observed. The prior supplies a source-constrained transformation direction while the recorded continuation remains the target; inversion supplies no new physical evidence. TTI forms training pairs; deployment uses raw history.

a) Preserving task evidence: The instruction identifies entities, attributes, relations, and conditional cues. Let $\mathcal{J}$

contain all valid text keys, and let $\mathcal{G} \subset \mathcal{J}$ contain the tokenizer positions overlapping the selected task-bearing spans, excluding special/function tokens. Frozen source attention is normalized over $\mathcal{J}$ before selecting $\mathcal{G}$:

$$\begin{aligned} P_{iu,j}^{bh} &= \text{softmax}_{j \in \mathcal{J}}(Q_{iu}^{bh} K_j^{bh\top} / \sqrt{d}), \\ a_{iu} &= \text{mean}_{b,h} \sum_{j \in \mathcal{G}} P_{iu,j}^{bh}, \\ w_g(i, u) &= \frac{a_{iu}}{\sum_{i',u'} m_{i'u'} a_{i'u'} + \epsilon}. \end{aligned} \quad (3)$$

The reported TTI models were trained using Eq. 3. We average heads in blocks $\{6, 12, 18, 24\}$ at source-probe noise $\sigma = 0.05$. The same blocks supply F_ψ : features are resampled to a common space-time grid, normalized, concatenated, and unit-normalized again. This readout includes local as well as semantic features; cue preservation is tested empirically. The RGB bridge is direct partial-latent decoding, $R(U) = D_{\text{Wan}}(U)$. For valid source locations m_{iu} , we use cosine distance $d(a, b) = 1 - a^\top b$ and minimize

$$\mathcal{L}_{\text{pres}} = \frac{\sum_{i,u} m_{iu} w_g(i, u) d(F_\psi(R(U))_{iu}, F_\psi(\mathcal{H}_t)_{iu})}{\sum_{i,u} m_{iu} w_g(i, u) + \epsilon}. \quad (4)$$

Source features and task weights are detached. Gradients pass through the frozen decoder and transformed-feature readout to U ; neither video prior parameters nor source references are optimized. The objective constrains observed evidence, not an unobserved task answer.

b) Preserving dynamics: Source feature correspondences, inspired by video-editing consistency [32], [31], define a source-confidence-filtered set $\mathcal{C}_{\text{src}}$. Mutual nearest-neighbor matches require valid endpoints, cosine similarity ≥ 0.8 , and cycle error ≤ 1 feature-grid pixel. For $r = (iu, jv)$, let $D_r(H) = F_\psi(H)_{iu} - F_\psi(H)_{jv}$. We match changes to the source:

$$\mathcal{L}_{\text{dyn}} = \frac{\sum_{r \in \mathcal{C}_{\text{src}}} q_r \|D_r(R(U)) - D_r(\mathcal{H}_t)\|_2^2}{\sum_{r \in \mathcal{C}_{\text{src}}} q_r + \epsilon}. \quad (5)$$

Here $q_r \in [0, 1]$ weights source-match confidence. The objective matches source changes rather than making neighboring frames identical. Feature matching is not a visibility oracle: unmatched events receive no dynamic constraint, a limitation distinct from the per-frame source loss.

c) Guided partial inverse: The frozen video prior uses the T2V path of Wan2.2-TI2V-5B [34], [35], without a clamped first frame. From $Z_0 = E_{\text{Wan}}(\mathcal{H}_t)$, Euler integration moves toward partial noise before a bounded correction:

$$\begin{aligned} U_{k+1} &= Z_k + (\sigma_{k+1} - \sigma_k) v_\psi(Z_k, \sigma_k; \ell), \\ Z_{k+1} &= U_{k+1} - \eta_k \text{Clip}_{b_k}(\nabla_U \mathcal{L}_T(U_{k+1})), \\ \mathcal{L}_T &= \mathcal{L}_{\text{pres}} + \lambda_{\text{dyn}} \mathcal{L}_{\text{dyn}}, \quad \lambda_{\text{dyn}} = 1. \end{aligned} \quad (6)$$

Support normalization removes dependence on the number of valid matches; it does not equalize loss scales or gradients. For unit features, $\mathcal{L}_{\text{pres}} \in [0, 2]$ and $\mathcal{L}_{\text{dyn}} \in [0, 16]$. The eight-step inverse ends at $\tau_{\text{inv}} = 0.1$, independently of policy flow time. With $\eta_k = 0.01$, $\text{Clip}_{b_k}(g) = g \min(1, b_k / (\|g\|_2 + \epsilon))$ and $b_k = 0.01 \|U_{k+1}\|_2$ bound each correction; prior inverse

steps are detached. This guided path is not an exact inverse. The decoded history $\tilde{\mathcal{H}}_t$ is re-encoded for the policy. A pair is retained only with finite values, at least 64 valid matches, $L_{\text{pres}} \leq 0.10$, and $L_{\text{dyn}} \leq 0.16$. Rejection disables transformed supervision, leaving the raw example. Fixed thresholds yield 82% accepted coverage in the main configuration; Eq. 11 defines its training weight.

C. Unified Active-View World-Action Modeling

a) View-aware history: We extend EasyWAM-Unified [6] without replacing its current-condition or joint future/action interfaces. A frozen video VAE encodes each camera history independently. Camera identity, measured pose, relative exposure time, and validity distinguish view changes from scene changes. A temporal resampler aggregates these tokens:

$$\begin{aligned} V_t^c &= E_V(\mathcal{H}_t^c) + e_{\text{cam}}^c + e_{\text{time}}^c + e_{\text{pose}}^c + e_{\text{valid}}^c, \\ B_t &= G_\phi(\{V_t^c\}_c, \ell, s_t, \kappa, \delta_{\leq t}). \end{aligned} \quad (7)$$

The fixed-size summary B_t supplies recent evidence and its camera context; current RGB retains a fine-grained clean-condition path for contact and small cues. The summary is recomputed from the available window, not treated as supervised object slots or an indefinitely persistent belief. A cue outside the window must be reacquired to re-enter the visual context. Task priors or robot state can still affect success, so out-of-window success alone is not evidence of longer visual memory.

History summaries pass through a DiT-width [36] projector and join the context *after* native text projection, retaining the pretrained language interface. A validity mask excludes padding and missing frames. Pose and time help distinguish image displacement caused by a head turn from a scene change.

b) Observation actions inside the joint generator: Write $A = (A^{\text{arm}}, A^{\text{head}})$. Head coordinates augment the native action vector before action projection; they are not produced by a separate gaze ranker. The model processes current clean tokens, noisy future-video tokens, and projected noisy action tokens with shared observed context. Separate output projections return video and action velocities:

$$(\hat{v}_y, \hat{v}_{\text{arm}}, \hat{v}_{\text{head}}) = f_\theta(Y_s, A_s, s; O_t, B_t, \ell, s_t), \quad (8)$$

where O_t is the current clean visual condition and s is flow time. Splitting arm and head outputs for loss normalization does not introduce separate experts or a new planning stage.

Both future and action tokens can read observed context and interact during joint denoising. Observed features cannot read noisy targets. This differs from the action-only inference mask of Fast-WAM [4]: we retain future/action coupling in the Unified model. No clean future image or clean action target is supplied as an inference condition; the same validity mask is used for every comparison.

c) Where active-view supervision comes from: A demonstration synchronizes head/arm commands with images acquired along their physical trajectory. Video supervision models the ensuing view; action supervision reproduces the

associated sensing and manipulation motion, conditioned on task context. RoboTwin-AV trajectories come from privileged collection metadata: task roles, manipulation phase, and currently visible surface geometry. These are not policy inputs. No optimal-viewpoint labels or information-gain objective are required. Windows preserve action–observation timing; fixed-camera crops cannot supply physical head-action supervision.

Future targets remain in their recorded camera streams: canonicalizing them into a stationary object view would remove the effect of head motion that the generator must learn. Demonstrations associate search with reacquisition, approach with object–hand coverage, and placement with object–goal framing; zero head motion represents stay. These are learned continuations, not a deployed phase controller.

Inversion changes the observed condition while keeping the recorded head–arm continuation and future outcome. Paired learning encourages consistent actions under nuisance changes, provided that the deciding cue survives. Genuinely different cue states do not share a target by construction.

D. Paired Learning and Deployment

Raw and transformed passes share weights and nonvisual metadata, using their respective current RGB/history. Encoding the original current-plus-future clip *once* gives the common Y_0 ; original actions give A_0 . Mixing transformed current with raw future before causal-VAE encoding would change the target and is excluded. For independent noise on linear flow paths [37], [38],

$$Y_s = (1 - s)Y_0 + s\epsilon_y, \quad A_s = (1 - s)A_0 + s\epsilon_a. \quad (9)$$

Paired passes share $Y_0, A_0, s, \epsilon_y, \epsilon_a$.

For each branch $b \in \{\text{raw}, \text{stem}\}$, video, arm, and head errors are normalized by their valid coordinates and equally weighted:

$$\mathcal{L}_{\text{FM}}^b = \mathcal{L}_y^b + \mathcal{L}_{\text{arm}}^b + \mathcal{L}_{\text{head}}^b, \quad (10)$$

with velocity targets $\epsilon_y - Y_0$ and $\epsilon_a - A_0$. Separate head normalization prevents its two coordinates being diluted by arm dimensions; absent actuators have no loss. Valid pairs additionally align action velocities:

$$\begin{aligned} \mathcal{L}_{\text{pair}} &= \|\hat{v}_a^{\text{raw}} - \hat{v}_a^{\text{stem}}\|_{\text{valid}}^2, \\ \mathcal{L}_{\text{WAM}} &= \frac{\mathcal{L}_{\text{FM}}^{\text{raw}} + m_{\text{pair}}\mathcal{L}_{\text{FM}}^{\text{stem}}}{1 + m_{\text{pair}}} + \lambda_{\text{pair}}m_{\text{pair}}\mathcal{L}_{\text{pair}}. \end{aligned} \quad (11)$$

We attempt pairing for 50% of eligible windows and set $m_{\text{pair}} = 1$ only for accepted transformations, with $\lambda_{\text{pair}} = 0.05$. Per-window normalization prevents a second pass from doubling its weight.

a) Raw-history inference: We jointly denoise one trajectory, execute a short head–arm prefix, and recompute context from newly exposed raw frames. There is no online inverter or candidate ranking. A common adapter enforces actuator limits and usable-observation timing. Old frames retain their true ages. Predicted future latents are auxiliary outputs, never a replacement for physical measurements; deployment can omit RGB decoding.

Sec. IV-C.0.c separates video co-training from retaining future tokens at inference.

IV. EXPERIMENTS

a) Experimental questions: We test (Q1) inversion and evidence preservation, (Q2) observation control, (Q3) video co-training, paired learning, and test-time generation, and (Q4) composed physical tasks.

A. Benchmarks and Experimental Setup

TAVIS [11] uses native Head/Hands data and 96 test episodes/task/seed/split; pose OOD perturbs end-effectors ($\sigma = 10$ cm) and neck yaw/pitch ($\sigma = 10^\circ$). RoboTwin-AV adds synchronized RGB and executable pan/tilt to all 50 RoboTwin 2.0 tasks [17]. The gaze collector uses instance masks, rendered surfaces, a 0.06-rad stay deadband, and waypoint search. Collector inputs are withheld; success follows native predicates. We use 100 successful training trajectories, 20 development trajectories/task, and 100 test episodes/task/seed/condition: clean, appearance, initial-head-pose, and compound; seeds 41/42/43. Splits precede windowing; failures/timeouts remain.

Figure 3 illustrates representative simulated and physical task sequences.

a) Baselines and additional benchmarks:

ACT [39]/Diffusion Policy [40] train per task; π_0 [3], Fast-WAM and EasyWAM-Unified train per suite. Reruns share RGB, head–arm interfaces, and native predicates; AV-ALOHA [9] ACT uses joint14+head2 control. Published scores are separate. LIBERO [41] and LIBERO-Plus [42] retain native camera/action interfaces.

b) Training and metrics:

We fine-tune DiT/action modules and the 16-token adapter; VAE, text encoder, and video prior are frozen. Inputs use nine 240×320 frames over 0.8 s. Controls match initialization/recipe except the named factor, with trajectory-consistent head settings. SR averages tasks then seeds; sample SD uses denominator two. Overlapping windows are not independent trials.

B. Main Benchmark Comparisons

a) Results on TAVIS: Gains over EasyWAM-Unified are 4–16 percentage points (pp; Table I): 12/11 pp for Head and 14/13 for Hands under pose shifts on GR1T2/Reachy2. Head ID mean gaze–action lead time is 0.75/0.78 s (97% valid among 878/763 successful episodes), measured from final pre-grasp neck fixation to grasp completion. Failures are excluded; longer is not inherently better.

b) Results on RoboTwin-AV: Table II shows gains of 3.3/14.7/8.0/18.3 pp above the strongest listed baseline per column; compound exceeds Fast-WAM by 20.3 pp. Raw/full compound SR is $41.7 \pm 1.5/53.3 \pm 1.5\%$.

c) Fixed-camera generalization: Native-camera gains (Table V) establish robustness, not active sensing.

C. Mechanism and Observation Controls

a) History inversion and evidence preservation:

Task/dynamic preservation ablations cost 12.3/7.3 pp compound SR (Table III). The framewise control (43.5%) matches prior/readout capacity, not native Stem-OB. Simpler controls also fall short. A blinded audit (1,000 readable and 1,000

TABLE I: Success on TAVIS (%). The first row reproduces the authors’ reported π_0 results. Our evaluations retain native demonstrations and sensor access and average task success over three training seeds, with 96 episodes/task/seed per condition. ID/S/P denote in-distribution, spatial OOD, and initial-pose OOD.

Method	Head / GR1T2			Head / Reachy2			Hands / GR1T2			Hands / Reachy2		
	ID	S	P	ID	S	P	ID	S	P	ID	S	P
π_0 (published) [11]	47.1	27.9	1.9	38.8	22.1	11.2	70.1	49.0	14.6	76.7	51.0	38.2
π_0 (rerun)	46.0	25.0	3.0	40.0	24.0	13.0	70.0	50.0	14.0	77.0	51.0	39.0
Fast-WAM	47.0	29.0	6.0	42.0	27.0	17.0	72.0	51.0	23.0	75.0	53.0	41.0
EasyWAM-Unified	51.0	30.0	12.0	45.0	26.0	20.0	71.0	53.0	24.0	75.0	53.0	40.0
ActiveWAM (ours)	61.0	46.0	24.0	53.0	42.0	31.0	77.0	64.0	38.0	81.0	64.0	54.0



Fig. 3: Representative head-view sequences with pan/tilt configurations: placing bread in a pan in RoboTwin-AV (top); pouring oil and egg mixture, then stirring on the real robot (bottom).

TABLE II: RoboTwin-AV SR (%): 50 tasks, three seeds, 100 episodes/task/seed/condition; shared sensor/action interfaces. App./Pose/Both: appearance/head-pose/compound shifts. Ours: mean with sample SD beneath (pp). Dashes: unreported conditions.

Method	Clean	App.	Pose	Both
ACT	41.0	20.0	30.0	13.0
Diffusion Policy	44.0	22.0	26.0	16.0
π_0	64.0	43.0	44.0	35.0
Fast-WAM	77.0	44.0	57.0	33.0
EasyWAM-Unified	75.0	46.0	56.0	34.0
AV-ALOHA (adapted)	—	—	—	13.0
ActiveWAM	80.3 ± 2.1	60.7 ± 0.6	65.0 ± 3.0	53.3 ± 1.5

unavailable-cue source windows) gives transformed accuracy/false positives of 96.5/3.5%, versus 98.0/3.0% for raw source. Annotator agreement is 95% (Cohen’s $\kappa = 0.85$ [43]); independent source labels define the reference, not prior features or acceptance scores. $\text{CueAcc} = N_{\text{correct}}/N_{\text{readable}}$, $\text{UnavailableFP} = N_{\text{false positive}}/N_{\text{unavailable}}$, $\text{Coverage} = N_{\text{accepted}}/N_{\text{attempted}}$, and $\text{EventMAE} = N_e^{-1} \sum_e |\hat{t}_e - t_e^*|$ over independently labelled, matched events (0.16 s). Unread-



Fig. 4: TAVIS prototype inversion: RGB and residuals against the $\tau = 0$ codec control at a fixed scale. This illustrates the VAE-latent prototype, not the final multi-scale prior or aggregate evidence.

able transformed cues are errors; event MAE is conditional on matching, not detection recall.

b) *Learned observation control*: Table IV’s 20-pp pose gain is a locked-to-learned system comparison. With matched grounding, learned versus relation look-at plus stay gives 63/61% SR, 88/81% co-visibility, and 1.90/2.50 rad travel. Randomized collection views/dwell give 53/42% full/raw compound SR; external head perturbations give 51% for full, probing collector dependence.

c) *Prediction and paired-learning controls*: Table VI crosses video loss and inference mode following Fast-WAM [4]. EasyWAM-Unified’s *infer_action* skips RGB decoding but retains joint video/action latent denoising,

TABLE III: RoboTwin-AV ablations (SR, %). Removals change only the named component; compound SD is 1.5 pp for full and raw history.

Variant	Clean	Compound
ActiveWAM (full)	80.3	53.3
w/o history	74.0	34.0
w/o inversion (raw history)	77.0	41.7
Same-Wan framewise inversion	80.0	43.5
w/o task preservation	79.0	41.0
w/o dynamic preservation	81.0	46.0
w/o pair consistency	82.0	46.0
w/o view metadata	78.0	46.0
Raw/raw equal-forward control	77.0	40.0
Task-safe augmentation	78.0	45.0
Task-weighted direct noising	78.0	46.0
VAE round trip	77.0	42.0
Source-feature auxiliary	79.0	45.0

TABLE VI: RoboTwin-AV attribution/sensitivity (SR, %): trajectory-consistent factorial data and matched training/inference graphs. Coverage: accepted/attempted inversion windows.

Inversion	Head	Clean	Compound
Off	Locked	74.0	25.0
On	Locked	77.0	30.0
Off	Active	77.0	41.7
On	Active	80.3	53.3
Video loss	Inference	Clean	Compound
On	Joint	80.3	53.3
Off	Joint	76.0	44.0
On	Action-only graph	81.0	53.0
Off	Action-only graph	76.0	44.0
λ_{dyn}	Coverage (%)	Clean	Compound
0	86	81.0	46.0
0.1	85	81.0	48.0
0.5	84	81.0	51.0
1	82	80.3	53.3
2	80	80.0	51.0

unlike the separately trained action-only graph. With ten evaluations and a four-action prefix, decode-off gives zero action difference and 53% compound SR. Video co-training helps both graphs. Equal-forward raw/raw and removal of pair consistency underperform full (Table III).

d) Interaction and sensitivity: Active-minus-locked inversion gains are 6.7 pp in compound and -0.3 pp in appearance (unrounded means). This system interaction changes physical trajectories, precluding fixed-input architectural causal attribution. Weight/decode scans change SR and coverage together. Direct/clean-estimate/partial-round-trip decoding gives clean SR 80.3/80/80%, compound 53.3/51/50%, and coverage 82/84/86%. The latter two decode clean estimates or latents denoised to zero; acceptance effects remain confounded.

D. Composed Real-Robot Tasks

AirbotPlay fixed-base dual arms use fixed-focal D455 RGB on an IQR pan/tilt head. Each of three task segments composes two separately trained policies: *slicing*, place cucumber horizontally and retrieve knife, then slice; *egg*

TABLE IV: Head control with inversion fixed. SR: clean/pose (%); Co-vis.: object-goal co-visibility (%); travel: rad/episode.

Head controller	SR	Co-vis.	Travel
Locked	77.0/45.0	61.0	0
Object tracking	77.0/58.0	74.0	4.10
Relation look-at + stay	80.0/61.0	81.0	2.50
Joint learned (ours)	80.3/65.0	87.0	1.70
+ matched grounding	81.0/63.0	88.0	1.90

TABLE V: Fixed-camera checks (SR, %). LIBERO-Plus weights all 10,030 variants by category counts; these are our evaluations.

Variant	LIBERO	LIBERO-Plus
Raw history	98.9	68.2
Task-safe augmentation	98.6	72.5
Source-feature auxiliary	98.7	72.0
ActiveWAM	98.8	76.8

TABLE VII: Physical kitchen successes: Stage 1/total out of the same 20 segment attempts (60/method). Total requires both subtasks.

Method	Cucumber	Egg	Mixture
$\pi_{0.5}$ [21]	8/2	10/7	12/11
Fast-WAM	7/1	11/6	11/7
EasyWAM-Unified	7/1	10/7	12/9
Ours w/o inversion	10/4	12/7	14/11
ActiveWAM (full)	11/7	15/11	14/12

frying, pour oil, then egg while the left arm takes the pan handle, then retrieve spatula and stir; *mixing*, pour cucumber then cooked egg, then retrieve spatula and stir-fry. Egg stirring retains the pan-handle grasp; stage switching is prescribed.

Human inter-segment actions enter neither evaluation nor montage. Table VII scores Stage 1/total on identical attempts: 20/task/method, 60/method, 300 overall. ActiveWAM’s Stage 1/total SR is 66.7/50.0%, versus 60.0/36.7% without inversion and 48.3/23.3% for Fast-WAM. ActiveWAM’s taskwise 95% Wilson intervals [44] are $[18.1, 56.7]/[34.2, 74.2]/[38.7, 78.1]\%$, assuming independent within-task attempts without session-cluster adjustment.

Failures before/after Stage 1 are 20/10 for full (30 total), versus 24/14 without inversion: stage counts, not causes. Commands use 20-Hz resampling/holding, not 20-Hz replanning.

V. CONCLUSION

ActiveWAM couples task-guided history inversion and observation control through shared continuation targets, supported by TAVIS, 50-task RoboTwin-AV, and composed kitchen tasks. Controls favor video co-training; omitting RGB decoding preserves actions while joint latent denoising remains. Deployment uses raw inputs, finite history, and prescribed stage switching without inversion.

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